

Monitoring Multi-Tenant AI Environments

Ensuring fair use, performance, and security with per-tenant visibility - the foundation for cost allocation, anomaly detection, and compliance



Why Monitoring Matters



Detect Noisy Neighbors

Identify tenants hogging GPU resources, causing performance degradation for other users



Fair Cost Attribution

Accurately allocate infrastructure costs across tenants based on actual usage



Security Monitoring

Catch suspicious activity like cross-tenant access attempts and maintain audit trails for GDPR, HIPAA, and SOC 2 compliance

Key Monitoring Dimensions

1

Compute

CPU, GPU, and memory utilization tracked per tenant

2

Storage

Datasets, model checkpoints, and feature stores usage metrics

3

Networking

Ingress/egress traffic patterns and cross-tenant communication attempts

4

Costs

Usage-based billing data for accurate chargeback

5

Security

Authentication failures, privilege escalation, and policy violations

Multi-Tenant Metrics Strategy

Per-Namespace

Kubernetes namespace-level metrics provide isolation and attribution

Per-Project

Cloud AI platforms metrics grouped by project boundaries

Per-Bucket

Storage metrics tracked at bucket granularity

Per-Service Account

Job-level granularity for detailed attribution



Tools for Multi-Tenant Monitoring

Prometheus

Open-source metrics collection system that efficiently gathers data from Kubernetes clusters and GPU resources

Grafana

Visualization platform for creating per-tenant dashboards with role-based access controls

Kubecost

Specialized solution for Kubernetes cost allocation, providing tenant-level resource usage and spending insights

ELK/EFK Stack

Centralized log monitoring with Elasticsearch, Logstash/Fluentd, and Kibana for searching and analyzing logs

Cloud-Native Solutions

AWS CloudWatch, GCP Cloud Monitoring, and Azure Monitor for integrated cloud service monitoring

GPU Monitoring with DCGM + Prometheus

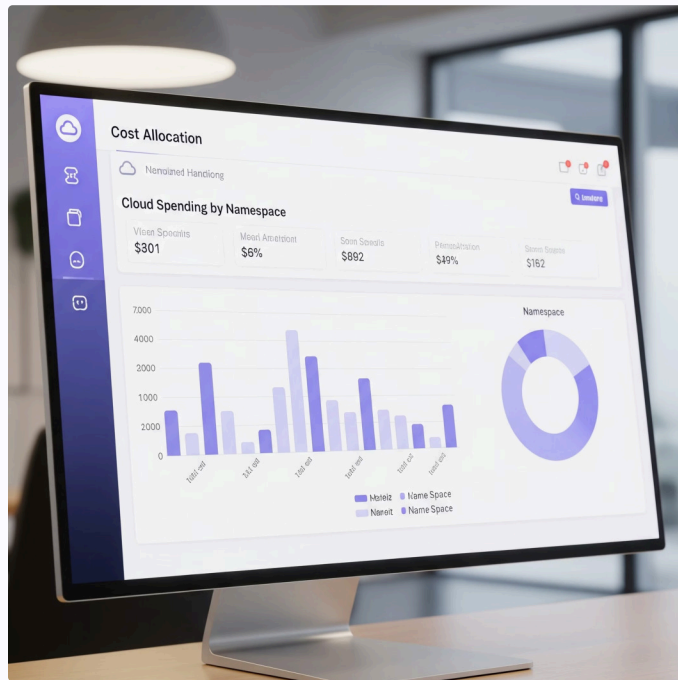
Implementation Strategy

- NVIDIA DCGM exporter collects detailed GPU metrics
- Prometheus scrapes metrics and applies tenant labels
- Grafana dashboards provide [per-namespace visualization](#)
- Detect resource hogging [before it impacts other tenants](#)



Real-time GPU metrics allow SREs to identify problematic workloads and ensure fair resource allocation.

Cost Monitoring with Kubecost



Key Benefits

- Integrates directly with Kubernetes clusters
- **Allocates infrastructure costs** based on CPU, memory, GPU and storage usage
- Enables accurate **chargeback/showback billing** for internal teams
- Provides granular visibility into spending patterns
- Eliminates disputes by giving each tenant visibility into their resource consumption

Security Monitoring in Multi-Tenant AI

Enable Comprehensive Audit Logs

- Kubernetes API server audit logs
- Cloud IAM access and authorization events
- Model registry and artifact access logs

❌ Security logs should be forwarded to a SIEM like Splunk, Datadog, or Azure Sentinel for correlation and advanced threat detection.

Critical Detection Scenarios

- Unauthorized API Access

Attempts to access resources outside tenant boundaries

- Cross-Namespcae Attempts

Workloads trying to access data from other tenants

- Abnormal Model Access

Unusual patterns in model downloads or inference requests



Alerting & Automation



Threshold Alerts

Trigger notifications when GPU utilization exceeds 90% per tenant or when memory consumption spikes unexpectedly



Cost Anomaly Detection

Alert on sudden increases in spending patterns that deviate from historical baselines for each tenant



Security Incident Alerts

Notify on suspicious patterns like repeated failed logins or unusual access patterns across tenant boundaries



Automated Responses

Implement automatic actions: throttle resource-intensive jobs, block suspicious users, or scale infrastructure

Multi-Tenant Monitoring Architecture



Comprehensive monitoring architecture enables visibility across tenants while maintaining proper isolation and access controls.

Challenges in Multi-Tenant Monitoring

Attribution Complexity

Shared services and infrastructure components are difficult to split fairly among tenants. Examples include control plane costs, shared databases, and gateway services.

Monitoring at Scale

Processing millions of metrics across thousands of jobs requires significant compute resources. High-cardinality metrics with tenant labels can overwhelm traditional monitoring systems.

Privacy Boundaries

Ensuring one tenant cannot view another tenant's metrics or logs requires careful RBAC implementation and potential data segregation at collection time.

Monitoring Overhead

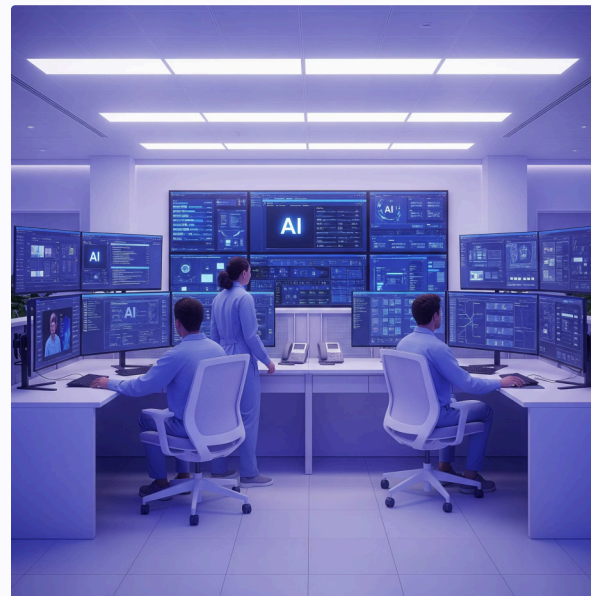
The monitoring infrastructure itself can consume 5-10% of cluster resources. This cost needs to be either shared fairly or absorbed as platform overhead.

Industry Monitoring Practices

Cloud AI Platforms

- **AWS SageMaker**: Provides per-user CloudWatch metrics and detailed billing integration
- **Google Vertex AI**: Implements per-project monitoring and logging with Cloud Monitoring
- **Databricks**: Offers per-workspace dashboards for compute and storage utilization

Enterprise AI Labs



Large enterprises typically combine Kubecost with Prometheus/Grafana for shared GPU clusters and implement custom dashboards for executive reporting.

Summary: Effective Multi-Tenant Monitoring

Key Takeaways

- Comprehensive monitoring is **essential for fairness, security, and compliance** in multi-tenant AI environments
- Track all five dimensions: compute, storage, network, costs, and security events
- Implement specialized tools: Prometheus, Grafana, Kubecost, and cloud logging services
- Per-tenant visibility builds trust and enables efficient resource utilization

📄 **Next steps:** Evaluate your current monitoring coverage, identify gaps in tenant isolation, and implement tenant-aware dashboards.

